BGIN Block #10 Meeting Report

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The Mutual Impact and Improved Outcomes of Blockchain & AI as Synergistic Technologies

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The Global Network for Blockchain Stakeholders™

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Block#10 Day2_9

Keynote Session Report: The Mutual Impact and Improved Outcomes of Blockchain & AI as Synergistic Technologies

Date: Monday, March 4, 2024

Location: Tokyo

Duration: Approx. 90 minutes

Summary

- The workshop opens with Speaker 1 discussing the integration of AI and blockchain, aiming to understand the services these technologies provide to each other, their inspirations, and the potential threats and implications for practitioners and policy. A working group will shape the research agenda based on the discussions. A discussant talks about the transformative potential of combining AI and blockchain, comparing it to past technological innovations like the iPhone and Netflix...

- The discussion continues with the discussant outlining future possibilities where AI and blockchain converge, including distributed model training and decentralized data curation. He introduces the concept of Borgs - entities integrating automation for dynamic DAOs. The moderator engages with his presentation, showing interest in Borgs and how they represent a cutting-edge development. The conversation opens to the floor, discussing AI's potential in writing smart contracts and the associated risks...

- Another discussant discusses their application enabling users to create AI chatbots tokenized as NFTs, highlighting potential revenue sharing through smart contracts. The discussion expands to cover various applications and challenges in integrating AI with blockchain, including the need for data privacy and transparency. The conversation then shifts towards regulatory and policy perspectives, expressing concerns about the rapid advancement of these technologies and the need for a balanced approach to innovation and regulation...

- Participants delve into discussions on the ethical use of AI and blockchain, emphasizing the importance of responsible innovation and the potential societal impacts. The need for multi-stakeholder collaboration is highlighted to ensure that the development of these technologies aligns with societal values and ethical principles.

- The conversation moves towards specific use cases and real-world applications of AI and blockchain integration, with a focus on the environmental, social, and governance (ESG)

sector. Another discussant presents a case where AI and blockchain are used to automate and streamline ESG compliance processes, enhancing data gathering, policy drafting, and scoring. The discussion emphasizes the importance of human oversight in validating AI-generated outputs and the role of blockchain in ensuring data integrity and transparency.

- Another discussant introduces Compass AI, a tool providing legal and regulatory guidance in the crypto space, highlighting the challenges of integrating AI into legal databases and the importance of user consent and privacy. The conversation shifts towards potential risks associated with AI and blockchain, including accountability issues and the difficulty in regulating decentralized systems. Participants express concerns about the potential for AI to exacerbate existing regulatory challenges and the need for new frameworks to address the unique risks posed by these technologies.

- Another discussant presents a structured view on the regulatory challenges posed by AI and blockchain convergence, highlighting the assumption of human accountability in existing regulatory frameworks and the potential for AI-enabled blockchains to circumvent or even remove human decision-making from the equation. He discusses the need for regulatory adaptation to address the rapid evolution of these technologies and the importance of incorporating AI into regulatory processes to effectively manage emerging risks.

- The discussion continues with an exploration of specific disruptions to regulatory governance caused by AI and blockchain, such as challenges to accountability, crisis resolution, and the hardcoding of regulations. A discussant suggests strategies for addressing these challenges, including the use of AI by regulators, granting governance bodies access to AI blockchain activities, and harmonizing regulations across jurisdictions. The conversation highlights the importance of proactive and adaptive regulatory approaches to manage the complex and evolving landscape of AI and blockchain technologies.

- Participants explore the distinction between AI challenging regulatory governance versus subverting regulation itself, the implications of AI-enabled blockchains for trading and credit decisioning, and the potential for regulatory nodes in blockchain networks. The conversation underscores the need for balancing innovation with regulation and the importance of preparing for unforeseen challenges in the rapidly evolving fields of AI and blockchain.

Transcript (anonymized):

03:58:10
Thank you all for joining today's session, focusing on the intersection of AI and blockchain within financial applications and social economics. Today's workshop aims to delve into the mutual enhancements and inspirations these emerging technologies offer each other and to discuss the potential threats and policy implications arising from their integration.

03:59:08
This topic is not only current but also pivotal for our future research direction. In addition to today's discussions, our working group, which meets every second Wednesday online, will further explore these themes and produce a comprehensive document based on our findings.

03:59:57
Today's session will leverage an AI tool developed by a participant's company to help synthesize our discussions. Although the session is recorded, we adhere to a policy where statements can be referenced without attributing them to specific individuals, to encourage open and dynamic dialogue.

04:00:37
We aim for an interactive session, balancing contributions from all participants. As we proceed, please consider how you might contribute, especially in the latter part of the workshop, which will focus on threats and frameworks. Additionally, we are looking for volunteers to assist with note-taking to capture the richness of our discussions.

04:01:32
Volunteers for note-taking are greatly appreciated, as their contributions are essential for documenting our insights. This human element is invaluable for our collaborative effort.

04:02:24
With the organizational details covered, let's introduce the experts joining us today. The first part of our session will cover the application of AI to blockchain technologies and vice versa. We're joined by specialists from diverse locations, including Dubai and Melbourne, bringing a wealth of knowledge to our discussions.

04:03:13
In the second part, we'll shift our focus to policy perspectives, hearing from additional experts who will share their insights on the implications of these technologies from a regulatory standpoint.

04:03:59

To kick off our discussions, one of our experts draws parallels between blockchain and AI and past technological revolutions, emphasizing the transformative potential of these technologies when combined. They highlight the importance of embracing innovation to avoid obsolescence.

04:04:54
Sharing their extensive experience in the crypto and web three spaces, another expert emphasizes the importance of demystifying these complex technologies to uncover practical opportunities for disruption and transformation.

04:05:46
Exploring the transformative potential of blockchain and AI begins with addressing common inquiries about current developments, distinguishing hype from genuine opportunities, and understanding the fundamental shift these technologies bring to trust and control. Blockchain's decentralization enables peer-to-peer exchanges without central oversight, which aligns with its design for disintermediation. Initially, AI's integration with blockchain aimed at establishing AI marketplaces, facilitating the exchange of AI-related assets through cryptoeconomics to foster global collaboration in AI advancement. However, these applications are just the beginning. The real synergy unfolds as AI reinforces blockchain's foundational principles and, in turn, blockchain enables new possibilities for AI applications.

04:06:58
Delving into practical examples, AI's capability to analyze blockchain transactions can enhance security and transparency, while blockchain's immutable records serve as a reliable dataset for AI to derive insights. Recent discussions at a leading blockchain conference showcased innovative ways AI and blockchain might merge, including distributed model training, decentralized data curation, and programmable rewards that leverage cryptoeconomics, introducing new paradigms for privacy, governance, and autonomous interactions within decentralized networks.

04:08:06
These emerging concepts suggest a future where decentralized governance and AI-driven optimization could reshape organizational structures and cooperation. For instance, Decentralized Autonomous Organizations (DAOs) could benefit from AI's ability to streamline decision-making processes and operational efficiency, transcending current limitations of manual coordination and enhancing flexibility.

04:09:11
The integration of AI extends to optimizing DAO functionalities, from smart contract adjustments to predictive governance, introducing the concept of legally incorporated entities that blend automation with DAO structures to focus on critical operations while freeing human

creativity. This synergy between AI and such entities accelerates the exploration of decentralized models, eliminating operational bottlenecks and enabling a focus on innovation.

04:10:21
In conclusion, the fusion of blockchain and AI not only facilitates unprecedented levels of cooperation but also imbues communities with the agility and insight needed for dynamic and responsive ecosystems, heralding a new era of decentralized innovation and human collaboration.

04:10:53
Expressing gratitude for the comprehensive overview, there's a keen interest in delving deeper into the discussed topics, especially those fresh from a prominent blockchain event. An inquiry is raised about the term "Borgs," highlighted as a novel concept, reflecting the cutting-edge nature of the subject matter when information is scarce online. The question seeks clarification on whether "Borgs" is related to or a derivative of "cyborgs," indicating the novelty and emerging significance of the term in this context.

04:11:41
The concept of "Borgs" is under discussion, with an inquiry about whether there is an established definition within the industry.

04:11:47
The explanation clarifies that "Borgs" is a term still in the process of being defined, deriving from "cyborgs". Unlike decentralized autonomous organizations (DAOs) which operate as singular entities, "Borgs" are envisioned as multiple, separate legal entities interacting with one another. AI plays a key role in facilitating this interaction, automating the communication and groundwork between these entities, thereby acting in a cyborg-like capacity.

04:13:50
A question arises about the capabilities of AI in the context of blockchain, specifically whether AI or blockchain code can autonomously write smart contracts. This inquiry touches on the advancements in AI that enable automated code generation in various programming languages, including Python.

04:14:22
The response addresses the cautious approach taken towards AI-generated code on immutable platforms like Ethereum, emphasizing the potential risks associated with deploying vulnerable or flawed code. While AI is increasingly used to assist developers, complete reliance on AI for coding, especially in blockchain, is not yet feasible. GitHub Copilot is mentioned as an example

of how AI currently serves more as a co-pilot to developers rather than assuming full control over the coding process.

04:15:02
A new question is prompted, indicating a continued interest in exploring the potential of AI in blockchain applications.

04:15:07
The inquiry delves into AI's ability to execute transaction flows autonomously, potentially eliminating the need for human intervention in signing off on transactions. This raises concerns about the implications for accountability and the possibility of reversing transactions in cases of errors or fraud.

04:15:36
The response highlights the emergence of AI agents conducting transactions with each other as a likely initial application, with automated payments introducing new risks. The discussion touches on the irreversible nature of blockchain transactions and the potential need for mechanisms like escrow to mitigate risks in automated transactions between AI agents.

04:16:26
Shifting the focus, a new question considers the possibility of leveraging blockchain to support AI development, rather than the converse. The idea of using blockchain for version control of AI models is proposed, suggesting that blockchain could provide a transparent and immutable record to track changes and identify when issues with AI models arise.

04:17:23
The potential for blockchain to offer a transparent and trustless system for AI data and development is acknowledged. The conversation considers how blockchain could make AI development more transparent, with version control on the blockchain providing clear insights into changes and developments in AI models.

04:18:04
The discussion transitions towards exploring specific applications of blockchain in supporting AI, with an introduction to upcoming insights on how AI assets can be stored and managed using blockchain technology, particularly in the form of Non-Fungible Tokens (NFTs).

04:18:37
A practical application is described where AI chatbots are tokenized as NFTs, allowing for innovative models of ownership and revenue sharing among developers through smart contracts. This approach facilitates direct compensation for developers based on user interactions with AI-

driven chatbots, showcasing a concrete example of how blockchain can support AI development and deployment.

04:19:55
A request for examples of adoption in the realm of tokenized AI assets and marketplaces prompts a discussion on similar initiatives within the industry.

04:20:01
The response acknowledges other entities in the space working on AI marketplaces and revenue sharing models, though noting the absence of widespread adoption thus far. The conversation suggests an industry interest in creating decentralized platforms for generative AI applications.

04:20:44
Inquiring about the characteristics of early adopters and suitable candidates for experimentation in the integration of AI with blockchain, the discussion seeks to identify the early movers in this space.

04:21:09
Clarification is sought on the question regarding the characteristics of early adopters in the space, leading to a discussion on the user base and applications of AI in web three environments.

04:21:23
The primary users of the discussed application are identified as game and anime enthusiasts, with a focus on providing companionship through AI chatbots. The discussion highlights a general curiosity about AI and its generative capabilities, particularly in creating illustrations and engaging with AI-driven chat services.

04:22:47
The conversation shifts to the integration of AI and blockchain for data learning, with a mention of Flare Network's unique approach to incorporating oracles directly into the blockchain, offering potential benefits for consensus machine learning and data integrity.

04:23:40
An invitation for further contributions to the conversation is extended, indicating a collaborative and explorative nature of the discussion. A request for repetition of Flare Network indicates continued interest in exploring its capabilities and implications for AI and blockchain integration.

04:23:54

The conversation reflects on the broader implications of integrating blockchain with AI, considering issues of transparency, explainability, and responsible AI use. The mention of principles such as those in the AI Bill of Rights and the role of blockchain in supporting these principles suggests a thoughtful consideration of how blockchain technology can contribute to the responsible deployment of AI.

04:24:49
The discussion touches on data management, privacy, and the potential for blockchain to contribute to the ethical and responsible use of AI. Concerns are raised about how well certain jurisdictions handle these issues and the importance of exploring how blockchain can address biases and unintended outcomes in AI applications.

04:25:23
The conversation shifts to the role of blockchain in fostering responsible AI adoption across different economies and jurisdictions, highlighting the interest of research institutions in leveraging blockchain as a tool to ensure ethical growth in AI usage.

04:26:23
A personal anecdote is shared about an experiment using AI to create a business from scratch, highlighting AI's extensive capabilities in automating various business processes, with the notable exception of tasks that require human identity verification, such as opening a bank account. This story underscores the vast potential of AI to automate and decentralize operations.

04:27:10
Reflections on the synergies between blockchain and AI raise concerns about the potential consequences of combining these technologies without responsible oversight. The conversation acknowledges the transformative power of each technology but also cautions against unbridled innovation that lacks ethical considerations.

04:27:57
The dialogue evolves into a broader contemplation of the implications of allowing AI and blockchain to interact freely, without regulatory boundaries, speculating on the potential outcomes of such a combination in the context of financial technologies.

04:28:22
The discussion concludes with a comparison to other potentially disruptive technologies, suggesting that the concerns associated with the convergence of blockchain and AI might not be as immediate or severe as those posed by other technologies, such as bioweapons, emphasizing the need to contextualize the risks and opportunities presented by these advancements.

04:28:31
In considering the integration of AI and blockchain, it's crucial not to overlook potential negative outcomes. This technological convergence, while offering immense potential, also presents a "Pandora's box" of challenges that, once opened, may be difficult to contain. It's essential for the tech community to ponder the ethical implications and ask "should we?" before proceeding, rather than merely "could we?" This sentiment is especially pertinent given the expertise in policy and regulation present here, prompting an early segue into discussions about potential threats and the need for careful policy consideration.

04:29:10
The conversation includes a noteworthy anecdote about an individual who, using AI, managed to automate the entire process of creating a business, from concept to design, highlighting the extensive capabilities of AI. The only task AI couldn't accomplish was opening a bank account, underscoring a current limitation in fully automating business operations through AI. This story raises questions about the future implications of combining AI's automation capabilities with blockchain's financial transaction functionalities, and whether this synthesis could lead to unforeseen consequences.

04:30:01
A practical example of AI and blockchain integration is the development by a team working on a scalability solution for a well-known cryptocurrency, enabling AI chatbots to autonomously handle transactions using a specific protocol. This innovation allows AI agents to autonomously pay for resources, showcasing the potential for AI to autonomously conduct financial activities and make decisions, leading to a significant number of microtransactions and decisions being made autonomously by AI agents every day.

04:31:01
The discussion transitions to focus on the financial applications of AI and blockchain, highlighting the importance of practical, real-world applications over speculative ideation. The emphasis is on implementations that enhance user experience and compliance processes in areas such as environmental and social governance, without leading to dystopian outcomes.

04:31:33
A clear distinction is made between numerical and semantic analysis in AI, with an emphasis on the latter, particularly large language models like ChatGPT, which are not sentient. These models are stateless semantic machines, meaning they don't retain information between interactions. This characteristic is crucial to understand when considering their application in various workflows, including those related to environmental and social governance compliance, highlighting the practical aspects of integrating AI and blockchain in real-world applications.

04:33:49
The comparison between large language models (LLMs) and blockchains is highlighted, noting the stateless nature of LLMs in contrast to blockchains, which function as state machines with memory. This distinction is emphasized as foundational to understanding the complementary roles these technologies can play in applications beyond dystopian scenarios, fostering a positive outlook towards their integration.

04:34:30
The discussion pivots to the practical adoption of blockchain, emphasizing the need for real-world use cases that significantly impact users' daily lives to drive broader acceptance. The narrative suggests that while blockchain may offer improvements over traditional systems, it might not present a drastic enhancement for the average user, unlike the compelling experiences offered by LLMs like ChatGPT.

04:35:28
The concept of using LLMs as a front-end technology and blockchain as a back-end technology is proposed as a strategy for facilitating wider blockchain adoption. This approach is posited as a means to leverage the intuitive and engaging user experiences enabled by LLMs to drive interest and engagement with blockchain applications.

04:36:31
An example is provided of automating and streamlining Environmental, Social, and Governance (ESG) data collection processes for customers, transitioning from a traditional survey approach to an LLM-based method. This shift allowed for the transformation of unstructured data into structured data, enhancing user engagement with the ESG information gathering process.

04:37:38
The introduction of a "microSIG" technology is discussed, enabling the creation of blockchain-based hashes of LLM outputs to facilitate human verification and attestation. This process is designed to insert a layer of human oversight and ownership into the use of LLMs, ensuring that LLMs do not autonomously make decisions without human intervention. The discussion also touches on the potential for storing ESG scores on a blockchain, highlighting the benefits of immutable and transparent record-keeping in the context of ESG compliance.

04:38:41
The financial implications of Environmental, Social, and Governance (ESG) scoring for companies are discussed, highlighting how such scores can affect loan terms with banks. The integration of Large Language Models (LLMs) and blockchain is credited for streamlining and enhancing the credibility of the ESG assessment process, potentially leading to the tokenization of green or ESG-focused investments. This approach could revolutionize asset management by

allowing for the creation of investment pools based on ESG criteria, facilitated by blockchain and decentralized finance (DeFi) technologies.

04:39:49
The potential for widespread blockchain use is linked to the development of such practical applications, which could significantly enhance the technology's adoption by demonstrating its tangible benefits. The conversation suggests that as more use cases emerge, blockchain's impact and utility will become more apparent to a broader audience, accelerating its integration into various sectors.

04:40:29
A call is made for additional examples of AI and Web 3.0 innovation as the discussion approaches the subject of potential challenges in integrating these technologies. This is aimed at enriching ongoing research and documentation efforts with practical insights.

04:41:06
A question is raised about the reliability of LLM outputs for ESG evaluation, particularly when dealing with unstructured or qualitative data. Concerns are expressed about the dependability of such automated assessments and the role of blockchain in ensuring the integrity of the resulting scores.

04:42:33
The conversation acknowledges the complexities involved in generating accurate LLM outputs, emphasizing the importance of human oversight in validating and utilizing AI-generated data. The discussion points to the need for LLMs to function as tools aiding human decision-makers, rather than making autonomous decisions, to ensure the responsible use of AI.

04:43:36
The challenges associated with AI, including issues of bias and unrealistic outputs, are acknowledged, highlighting the limitations and imperfections of current AI tools. An anecdote illustrates the potential societal impact of these limitations, particularly in shaping perceptions and expectations among young women. The discussion underscores the nascent state of AI technologies and the importance of critical engagement and oversight in their development and application.

04:44:15
As we delve into the second part of our workshop, we focus on the inherent risks associated with nascent technologies. Acknowledging that the early stages of technology development often involve chaos and mistakes, the discussion aims to identify these risks and explore potential frameworks for understanding and eventually mitigating them. The objective is not necessarily to

find immediate solutions but to develop a language and approach for comprehensively understanding the risks involved.

04:44:53
Introducing a practitioner with a wealth of experience in the legal aspects of cryptocurrency, the conversation opens up to insights from both a legal and practical standpoint. With extensive experience advising on various aspects of crypto, from ICOs to NFTs and DeFi, the speaker brings a global perspective to the discussion, informed by a comprehensive database of crypto-related legal and regulatory knowledge worldwide.

04:45:51
The speaker shares their venture into utilizing AI to enhance their ability to provide high-quality legal advice and navigate a vast database of global crypto knowledge. They introduce Compass, a chat tool designed to offer navigational aid in the complex landscape of crypto legal and policy matters. Despite initial expectations of ease due to the proliferation of AI tools and open-source models, the actual development process presented significant challenges, emphasizing the complexity and friction involved in creating user-centric AI applications.

04:46:52
The speaker highlights the user experience challenges faced during the development of Compass, emphasizing the importance of user consent and involvement in design decisions. By adjusting AI model settings to balance creativity with factual accuracy, the tool aims to minimize "hallucinations" or inaccuracies while maintaining user agency in the interaction. Users are given the option to choose between different AI models, reflecting the dynamic nature of AI development and the importance of accommodating user preferences and needs.

04:47:51
Further discussing the design choices in Compass, the speaker notes the trade-offs between reducing AI "hallucinations" and maintaining model creativity. Users are given the ability to toggle between different AI models, including OpenAI's API and a self-hosted version, with plans to support additional models in the future. This approach allows users to personalize their experience and reflects the ongoing evolution of AI technology and understanding.

04:49:13
The conversation shifts to a participant expressing interest in Joni's approach to leveraging a substantial database for AI-driven legal and regulatory guidance in the crypto space. They share their own experience with a collection of blockchain white papers, probing into the scale of Joni's database and whether its size poses any technical constraints.

04:49:43
In response, the details of managing a large knowledge base are discussed, highlighting the use of vector databases and retrieval methods to enhance accuracy and response speed. The conversation touches upon the challenges of hosting and processing data, including geographical limitations and the impact on processing power and costs.

04:50:44
A gentle reminder is given to focus the conversation on broader topics rather than specific business strategies, steering the discussion back towards the workshop's main themes.

04:51:39
The dialogue turns to the critical aspects of transparency and explainability in emerging technologies, especially when considering the combination of AI and blockchain. The unique challenges posed by decentralization, such as delineating clear lines of responsibility and accountability, are brought into focus.

04:52:32
Drawing on personal experience in defense and policy, the speaker elaborates on the regulatory considerations for AI in the context of dual-use technologies. The conversation delves into the potential risks associated with AI capabilities that could be leveraged for both commercial innovation and more nefarious purposes, such as weapons development and cyber operations.

04:53:21
Concerns are raised about the implications of deploying AI on decentralized networks, particularly the challenges in controlling or mitigating risks associated with highly decentralized and autonomous systems. The conversation underscores the complexities of ensuring security and accountability in such scenarios, reflecting on a hypothetical interaction with an AI to illustrate these concerns.

04:54:09
The conversation highlights the increased risk of cybercrime due to the widespread availability of advanced AI technologies like deepfakes, which have undermined traditional defense mechanisms against scams such as business email compromise fraud. The discussion suggests that sophisticated AI and blockchain might offer solutions by enabling AIs to identify and report malicious activities, potentially through incentivization mechanisms.

04:54:59
The primary concern is articulated as the combination of AI's sophisticated capabilities with the decentralized and ubiquitous nature of blockchain, which complicates efforts to control or remove harmful content and holds entities accountable for their actions.

04:55:14
A participant suggests that the real issue might not be the combination of blockchain and AI per se, but rather the accessibility of open-source AI models that can be downloaded and run by anyone, potentially leading to their use for malicious purposes without the need for decentralization.

04:55:51
The discussion acknowledges that while individual malicious actors can be addressed, the challenge escalates when harmful AI applications are redundantly deployed across decentralized networks, making them difficult to remove and control the damage they could cause in a short period.

04:56:41
Despite recognizing these significant risks, the conversation notes that not all advancements should be halted due to potential misuse. The discussion touches on regulatory developments in the US that could have profound implications for decentralized finance (DeFi) and the broader technology landscape, advocating for a balanced approach to innovation and regulation.

04:57:27
As the discussion opens up for more input, a participant defers their turn, indicating they believe their points have been covered, thereby passing the opportunity to another participant to share their thoughts on the complex interplay between AI, blockchain, and the regulatory environment.

04:57:40
The conversation continues with a participant expressing optimism about the future of AI, emphasizing its potential as a remarkable tool. The discussion acknowledges that the advancements in AI are irreversible, suggesting that society must adapt to these changes rather than attempting to revert to previous states. The idea is proposed that AI could serve humanity by monitoring activities and providing necessary interventions where human oversight is required, especially given the vast amounts of data beyond human capacity to manage.

04:58:49
The notion is reinforced that the "genie is out of the bottle" concerning AI technology, indicating a point of no return. The conversation shifts towards leveraging AI to assist humans, highlighting the possibility of employing AI to monitor other AIs and human actions, ensuring checks and balances in the system.

04:59:41

The discussion touches on the challenges and fears associated with AI, suggesting the development of mechanisms where AI can serve as a safeguard, monitoring activities and signaling when human intervention is necessary. This approach acknowledges the limitations of human and regulatory capabilities in managing the complexities introduced by AI and decentralized technologies.

05:00:41
The potential benefits of AI in the context of decentralized communities, DAOs, and DeFi protocols are explored. The conversation highlights how AI tools could support these communities in managing operational risks and adhering to predefined rules and regulations, offering a valuable resource to sectors traditionally less focused on risk management.

05:01:53
The discussion concludes with a positive outlook on the intersection of AI, Web 3.0 protocols, and financial services, suggesting that the integration of these technologies could lead to beneficial outcomes for the broader community.

05:02:08
A participant is invited to share their insights, with the session moving towards a presentation segment. The plan is to allocate time for both the presentation and subsequent discussion, aiming to enrich the workshop with diverse perspectives and deepen the exploration of the topics at hand.

05:02:48
The conversation shifts towards a presentation focusing on preparedness for the regulatory response to the emergence of AI and blockchain technologies. Drawing on research and experience in blockchain governance, including work with MIT Sloan and involvement in developing blockchain standards, the presenter aims to delve into the intersection of AI and blockchain governance. This perspective is informed by years of regulatory experience and a broad range of interviews conducted across various jurisdictions and sectors.

05:03:29
Reflecting on a background that spans roles such as CIO at a central bank and participation in financial regulatory councils, the discussion emphasizes a long-standing engagement with blockchain from a regulatory viewpoint. The upcoming insights are based on extensive research and interactions with professionals in the field, including some participants in the current session.

05:04:13

The relevance of AI to blockchain is outlined, highlighting AI's potential to augment human capabilities through computational power and its ability to generate new knowledge and information. The discussion touches on the extensive research and development in generative AI, noting a shift towards improving computational efficiency, which could lower costs and expand deployment possibilities.

05:04:52
A role at a major tech company is mentioned, shedding light on the significant investment in generative AI, particularly in making it more computationally efficient. This advancement is expected to broaden the use and application of AI beyond high-cost server environments, making it accessible for various uses.

05:05:36
The complementary nature of AI and blockchain is further explored, emphasizing how each technology can address the other's limitations. Blockchain's capacity for ensuring integrity and providing version control is seen as beneficial for AI's need for data provenance and explainability. Conversely, the extensive data captured by blockchain can enhance AI's performance, suggesting a synergistic relationship between the two technologies. The presentation suggests that, despite potential doomsday scenarios, a more nuanced examination of how these technologies can mutually benefit each other is warranted.

05:06:28
The dialogue expands to consider the broader digital convergence encompassing virtual reality, robotics, and advanced battery technologies alongside AI and blockchain. This convergence is anticipated to drive significant, possibly unforeseen, innovations and disruptions in the near future. Reflecting on time spent in Washington D.C., the importance of viewing current threats and capabilities through various lenses to project future scenarios is emphasized, suggesting a range of potential outcomes from the amalgamation of these technologies.

05:07:10
Rather than focusing on specific details, the approach is to apply different perspectives, such as deepening, broadening, and accelerating, to existing threats to envision future scenarios. This method aims to prepare for and manage potential risks arising from the convergence of digital technologies.

05:07:56
The conversation shifts towards the regulatory challenges presented by the convergence of AI and blockchain. Regulation, traditionally based on the premise of human accountability, faces significant challenges in adapting to the decentralized and automated nature of these technologies. The discussion highlights how current regulatory frameworks, deeply ingrained

with the assumption of human responsibility, may struggle to address the complexities introduced by AI and blockchain integration.

05:08:43
The inherent assumption of human accountability in regulation is critically examined, noting how AI already challenges this premise by potentially circumventing existing controls, such as de-anonymizing data in ways that conflict with privacy regulations like the GDPR.

05:09:27
The combination of AI and blockchain is identified as a particularly potent challenge for regulatory governance. AI's potential to enable blockchain applications further complicates regulatory efforts, especially when AI-enhanced blockchains could obscure or eliminate human accountability, raising questions about the effectiveness of current regulatory approaches.

05:10:21
The discussion touches on the implications of AI-enabled blockchains for regulatory enforcement, particularly in scenarios where autonomous decisions are made without clear human oversight, such as in decentralized autonomous organizations (DAOs). The conversation acknowledges the limitations of relying on the "human in the loop" concept, especially when individuals or entities may intentionally seek to operate outside regulatory frameworks.

05:11:07
And my first response is, when that happens, you're at real risk of your first contact with something like generative AI being the victim of it. If someone is maliciously using it against you or your organization, and you have not equipped yourself with knowledge, let alone defenses against it, because you've decided to watch and see, it means those who are not worried or actually actively trying to circumvent regulatory controls, can actually use the weaknesses we've talked about very consciously, and architect them into what they're doing to completely attack your organization. And if your first contact is in a defense context, you're in big trouble. One thing I will, and it's interesting. We talked about this probably only about 20 minutes ago.

05:11:50
There is a real wargame scenario where an AI enabled blockchain has not only access to significant on chain data and computational resources, what we know is data and computational resources. What allows AI to improve itself, feed it more data, and feed it more computational capability, and an LNM or foundation model of any kind can actually tune itself and actually reweight itself and improve its power. Where it gets particularly interesting is if something which has the ability to use on chain data to improve itself also has the ability to have non financially regulated value.

05:12:29
And whether that's non regulated cryptocurrencies or on chain currencies, which aren't standard financial fiat currencies, the ability to use that value on chain to actually accumulate more data and computational capability creates the ability for something to potentially improve itself at a rate that exceeds human regulators ability to keep up. And so there is the scenario where, again, you can regulated entities, and generally they play well. The danger is when someone consciously wants to subvert regulation and actually create a capability that bypasses it. Certainly, I think right now, it'd be a lot easier to create and construct if you were to play a mental game to say, how would I bypass regulation and create a scenario which demonstrates number five in action?

05:13:21
It wouldn't take too many geniuses to actually war game that and come up with any number of scenarios where that would be quite effective in the current regime. I published a paper on this about two years ago at ISIS in Copenhagen, and it was actually seen to be very far fetched. We were surprised it actually got accepted. But these days, I think it's not science fiction at all. It's very plausible.

05:13:49
This is just an example of that. On the left, you see a current regulated regime where you have institutions and individuals, and again, a lot of regulatory controls targeted at boards and leadership. But if on the right, we have a scenario where you have an AI enabled blockchain, if you actually start making decisions and having individuals interact with capabilities that are no longer subject to board or leadership oversight, a big percentage of our existing regulatory regimes start to fail. And so I've only got two slides to go, and then I'll wrap. These are four disruptions that we've sort of seen in the near horizon, some of which are already here, which are very relevant for existing regulatory and governance regimes.

05:14:33
One is the first one we've already covered, which is that assumptions around the ability to target accountability at individuals can very quickly be displaced by AI enabled blockchains, particularly if you are intentionally attempting to displace them. Secondly, we don't often talk about the crisis scenario. I worked in the central bank for a decade, and what you find is often there are key entities whose responsibilities are not specified or programmed, but they have the role of fixing a crisis when it occurs, and they are trusted to fix those cris. The more activity is moved on chain and the more AI is involved in decision making. Normally, those kind of AI capabilities are not trained around crisis resolution. They're trained around optimization and performing the bulk of actions in a governed state. They're very rarely trained on crisis resolution, which typically are unplanned and very complex.

05:15:26
Exploring the challenges regulators face with the rise of AI and blockchain, there's concern about crisis management when control is increasingly delegated to autonomous agents. The ability for these agents to effectively address unforeseen crises, typically handled by human intervention, poses a significant challenge, especially in a regulated context.

05:16:12
The discussion then turns to the legislative and regulatory frameworks, which are deeply ingrained with assumptions of human accountability. The monumental task of revising vast amounts of legislation, built on centuries of human-centric legal principles, to accommodate the nuances of AI and blockchain is highlighted.

05:16:54
Considering the rapid evolution and action of AI-enabled blockchains, the conversation acknowledges the potential mismatch between the pace of technological advancement and the regulatory response, emphasizing the need for innovative regulatory approaches.

05:17:45
Proposing potential solutions, the discussion suggests incorporating AI into regulatory processes for more effective oversight and risk management. The idea of regulatory bodies having direct access to blockchain activities (regulatory node access) and the importance of harmonizing regulations across jurisdictions to avoid regulatory gaps are also discussed.

05:18:11
A participant raises a question about the distinction between AI challenging regulatory governance by subverting compliance technologies and directly subverting regulation itself, seeking clarity on the implications of these challenges for regulatory frameworks.

05:19:35
The discussion acknowledges that the introduction of AI has compounded the challenges initially anticipated by privacy-focused regulations like GDPR, which were designed to address how organizations manage customer and personnel information. The conversation emphasizes that the convergence of AI and blockchain adds even more complexity, suggesting that the challenges are cumulative, building on top of each other as these technologies evolve and intersect.

05:20:09
A participant draws a parallel between the current discussion on AI and blockchain and the historical transition in institutional trading desks from human traders to algorithmic trading over the first decade of the 2000s. This transition, which significantly reduced the need for human

traders, is presented as an example of governance adapting to technological advancements without leading to widespread instability in financial markets.

05:21:11
In response, it's highlighted that regulators have already started to address the challenges posed by AI in areas like credit decisioning in banking by enforcing requirements around explainability, provability, and accountability. The conversation suggests that similar regulatory mechanisms could be applied to the integration of AI and blockchain in regulated contexts to mitigate risks and maintain control.

05:22:06
The discussion shifts to consider the potential for entities to intentionally bypass regulatory controls, emphasizing that while regulated entities might adopt AI and blockchain innovations within the boundaries of regulation, there's a concern about those who might seek to exploit these technologies for illicit purposes, outside of regulatory oversight.

05:22:49
The conversation concludes with an acknowledgment that regulators must consider both entities seeking competitive advantages within regulatory frameworks and those aiming to circumvent regulations altogether. The latter scenario raises concerns for law enforcement and broader regulatory bodies beyond financial regulators, highlighting the need for vigilance against the misuse of AI and blockchain by criminal organizations and black market operators.

05:23:38
A question is raised about whether regulators might be disincentivized from pursuing legislative reform due to the slow and cumbersome nature of such processes. It's suggested that regulatory bodies might prefer to extend their control to adapt more swiftly to real-time changes, potentially increasing their powers in a fast-evolving environment.

05:24:26
The response acknowledges the complexity of the issue, noting that regulatory expansion might be the most straightforward path to coverage but comes with its own challenges. It's pointed out that significant legislative changes often require a crisis to motivate action, suggesting that the urgency for AI and blockchain regulation might arise from societal harm or financial losses.

05:25:13
It's acknowledged that all three branches of governance (regulatory, legislative, and judiciary) are currently navigating the challenges posed by emerging technologies, but significant action typically follows societal harm, as seen in the cryptocurrency space.

05:25:47
Another participant is prompted to keep their comment brief due to time constraints.

05:26:01
A comment is made on the potential for emerging technologies to provide regulators with additional tools for enforcement, suggesting that the technology could arm both regulators and those being regulated, potentially leading to a more balanced and effective regulatory environment.

05:26:54
The conversation concludes with a discussion on the need for regulatory controls to evolve alongside technological advancements, particularly in the context of decentralized systems. The possibility of redistributing regulatory control points to include autonomous agents and decentralized entities is explored, suggesting a future where enforcement mechanisms and frameworks might also need to be open-sourced and decentralized.

05:27:42
The discussion explores the potential for regulatory bodies to leverage the underlying logic and frameworks of emerging technologies, even when data privacy is maintained. This approach could provide regulators with powerful tools to understand and monitor technological operations, enhancing transparency and regulatory oversight.

05:28:24
The session concludes with an acknowledgment of the valuable insights gained and an invitation for continued dialogue. Participants interested in further discussions are encouraged to connect with speakers through the workshop's organizers.

05:29:07
The next steps involve testing the views expressed during the workshop against further research, organizing ideas into a structured narrative to communicate current thoughts on the intersection of these technologies to stakeholders. The forum aims to identify emerging principles and highlight areas needing more exploration, inviting participants to join fortnightly online discussions to delve deeper into these topics.